



University of Colombo, Sri Lanka

University of Colombo School of Computing

Data Structures and Algorithms II

Semester II

Two (02) Hours

Answer All Questions

Instructions to Candidates

1. The medium of instruction and questions is in English.
2. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
3. MCQ should be marked on the answer sheet and the essay question on the paper provided.
4. Write your **index number** on the answer sheet.
5. This paper has two (02) parts, Part I and II. **Part I** consists of **thirty-five (35) MCQ** questions for **seventy (70) marks**, and **Part II** consists of an **essay question** for **thirty marks (30)**.
6. Calculators, Mobile Phones or any electronic devices are **not** allowed.
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******* Multiple Choice Questions Marking Method *******

- Each MCQ question has five (05) answers with one (01) or more (up to 05) correct answers. Each MCQ question is worth two (02) marks. Suppose a question has two (02) correct answers and three (correct) answers, *+2 marks will be shared between the correct options (1 mark per option), and -2 marks will be shared between incorrect options (-0.67) per option*. If one correct and incorrect answer are marked, the final mark for the question will be $1 + (-0.67) = 0.33$.
- Each question can have a positive or a negative value as the final mark. **Both negative and positive values would be added together to calculate the mark for the MCQ section**. If a student scores less than zero (0) for the MCQ section (Part I), zero (0) would be considered as the mark for that section (Part I).

PART I

1. Which of the following asymptotic complexities for a given function $f(n)$ is/are true? Consider k as a constant where necessary.
 - (a) $(n + k)^m$ is $\Theta(n^m)$
 - (b) 2^{n+1} is $O(2^n)$
 - (c) 2^{2n} is $O(2^n)$
 - (d) $\sqrt{\log n}$ is $O(\log \log n)$
 - (e) $n^{\log n}$ is $O(2^n)$

2. Consider $f(n) = n^{\log n}$ and $g(n) = 2^{\sqrt{n}}$. Which of the following is/are true about $f(n)$ and $g(n) \forall n \geq 1$?
 - (a) $f(n)$ is greater than $g(n)$
 - (b) $g(n)$ is greater than $f(n)$
 - (c) Both $f(n)$ and $g(n)$ can be bound by $O(n^2)$
 - (d) Both $f(n)$ and $g(n)$ can be bound by $\Omega(\log n)$
 - (e) $f(n)$ is $O(\log \log n)$

3. Consider $f(n) = 2n + 3$. Which of the following is/are true about $f(n) \forall n \geq 1$?
 - (a) $f(n)$ is $O(\sqrt{n})$
 - (b) $f(n)$ is $\Omega(n \log n)$
 - (c) $f(n)$ is $\Theta(n)$
 - (d) $f(n) \leq n^3$ for $n \geq 2$
 - (e) $f(n) \leq n$ for $n \geq 2$

4. Consider $f(n) = n!$ and $g(n) = \log n!$. Which of the following is/are true about $f(n)$ and $g(n) \forall n \geq 1$?
 - (a) Both $f(n)$ and $g(n)$ are $\Omega(1)$
 - (b) Both $f(n)$ and $g(n)$ are $O(n \log n)$
 - (c) Both $f(n)$ and $g(n)$ has no tight bound
 - (d) $f(n)$ is $O(n^n)$
 - (e) $g(n)$ is $O(2^n)$

PART I

5. Which of the following is/are true about asymptotic notation for functions $f(n)$ and $g(n)$?
- (a) if $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$ then $f(n)$ is $\Theta(h(n))$
 - (b) if $f(n)$ is given then $f(n)$ is $\Theta(f(n))$
 - (c) if $f(n)$ is $O(g(n))$ then $a \cdot f(n)$ is $O(g(n))$
 - (d) if $f(n)$ is $\Theta(g(n))$ then $g(n)$ is $\Theta(f(n))$
 - (e) if $f(n)$ is $O(g(n))$ then $g(n)$ is $\Omega(f(n))$

Consider the **Graph (G)** in Figure 1 for **Questions 07 to 10**

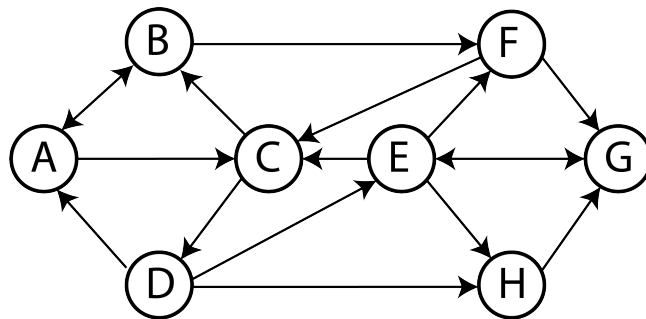


Figure 1: Graph for Questions 07 to 10

6. Which of the following paths is/are circuits?
- (a) A - B - F - C - B - A
 - (b) A - B - F - C - D - A
 - (c) C - D - E - F - G - E
 - (d) D - E - F - G - E - H
 - (e) H - G - E - C - D - H
7. Which of the following paths is/are cycles?
- (a) A - B - F - C - B - A
 - (b) A - B - F - C - D - A
 - (c) A - C - D - H - G - E
 - (d) C - D - E - F - G - E
 - (e) D - E - F - G - E - H

PART I

8. Which of the following is/are true about graph G in Figure 1?
- (a) $A - B - F - G$ is a simple path
 - (b) $A - C - B - A$ is a simple path
 - (c) $A - B - F - C - B - A$ is a Hamiltonian cycle
 - (d) E is a pendant vertex
 - (e) Shortest length from vertex A to G is three (03)
9. Which of the following is/are true about graph G in Figure 1?
- (a) $\deg^-(A)$ is 2
 - (b) $\deg^-(C)$ is 2
 - (c) $\deg^+(C)$ is 3
 - (d) $\deg(G)$ is 0
 - (e) $N(E)$ is $\{C, F, G, H\}$
10. Which of the following is/are true regarding a graph $G = (V, E)$ with V vertices and E edges?
- (a) If graph G is undirected, E is half the sum of the degree of each vertex V .
 - (b) If graph G is undirected, it has an odd number of vertices in an odd degree.
 - (c) If graph G is undirected, it has an even number of vertices in an odd degree
 - (d) If graph G is directed, the E equals the sum of the out-degree of each vertex V .
 - (e) If graph G is directed, the E equals the sum of each vertex V 's in-degree and out-degree.
11. Which of the following definitions defines a forest?
- (a) A disjoint union of trees.
 - (b) A tree that contains all vertices of a graph.
 - (c) A graph that has weights associated with each edge.
 - (d) A hierarchical structure that has no cycles.
 - (e) A graph that has no vertices.

PART I

12. Which of the following statements is/are true about the Adjacency Matrix implementation of graphs?
- (a) Can dynamically adapt to changes in the number of vertices in the graph.
 - (b) Requires $\Theta(V^2)$ memory.
 - (c) The matrix is dependent on the number of edges in the graph.
 - (d) The time to determine an edge exists in the graph is $\Theta(1)$.
 - (e) It takes $\Theta(V)$ time to list all vertices adjacent to a vertex V in the graph.
13. Which of the following statements is/are true about the Adjacency List implementation of graphs?
- (a) Can dynamically adapt to changes in the number of vertices in the graph.
 - (b) Requires $\Theta(V + E)$ memory.
 - (c) Can determine all the vertices adjacent to another vertex in $\Theta(\deg(V))$ time.
 - (d) The time to determine an edge exists in the graph is $\Theta(1)$.
 - (e) The best-case time to determine if an edge exists in the between vertices in the graph is $\Theta(V)$.
14. How many edges are in a K_6 complete graph?
- (a) 3
 - (b) 6
 - (c) 10
 - (d) 15
 - (e) 21
15. Which of the following statements is/are true about cyclic graphs?
- (a) All edges of the graphs are part of a cycle in a cyclic graph.
 - (b) Cycle graph and a cyclic graph refer to the same graph.
 - (c) Cyclic graphs have only one cycle.
 - (d) Cyclic graphs at least have one cycle.
 - (e) Disconnected graphs cannot be cyclic graphs.

PART I

16. What is an Euler Graph?

- (a) A graph that has no parallel edges except self-loops.
- (b) A graph that has an even degree for all vertices.
- (c) A graph that has a circuit with non-repeating vertices.
- (d) A graph that does not have a self-loop but has parallel edges between vertices.
- (e) A graph that is drawn in a single plane without two edges intersecting.

17. Which of the following is/are true about isomorphic graphs?

- (a) The graphs have an equal number of vertices.
- (b) The graphs have an equal number of edges.
- (c) Edge connectivity between graphs is preserved.
- (d) Has a bijection relationship between the graphs.
- (e) Euler graphs are isomorphic.

18. Consider a Binary Search Tree (BST) data structure with n nodes. Which of the following is always true regarding BST search?

- (a) The best-case time for BST is $\Theta(1)$
- (b) The best-case time for BST is $O(1)$
- (c) The worst-case time is $\Theta(\log n)$
- (d) The worst-case time is $\Theta(n)$
- (e) There is no tight bound for BST search in the worst-case

19. What is the operation represented by OperationX in the following code snippet?

```
int OperationX(int num)
{
    int A = 1;
    int B = 0;

    while (num)
    {
        B = B + A;
        A = A + 2;
        num = num - 1;
    }

    return B;
}
```

PART I

- (a) Decrement Operation
- (b) Log Operation
- (c) Increment Operation
- (d) Square Operation
- (e) Scale by a fixed number

Consider the following algorithm for classifying the edges of the graph in Figure 35 based on **Depth First Traversal** (DFT).

- **STEP 1:** If v is visited for the first time as DFT traverses the edge (u, v) , then the edge is a **tree edge**.
- **STEP 2:** Else, v has already been visited:
 - i If v is an ancestor of u , then edge (u, v) is a **back edge**.
 - ii Else, if v is a descendant of u , then edge (u, v) is a **forward edge**.
 - iii Else, if v is neither an ancestor or descendant of u , then edge (u, v) is a **cross edge**.

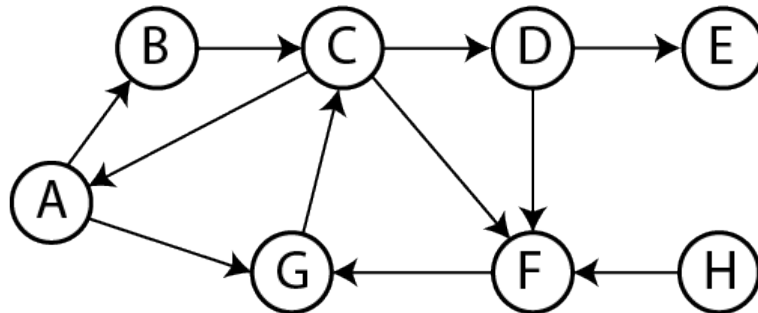


Figure 2: Graph for Questions 20 to 25

When processing the graph in Figure 35 to answer **Questions 20 and 25**, starting node of the DFT is **A**, and **adjacent edges** are chosen in **alphabetical order first**.

20. Which of the following sets contain only tree edges?
- (a) $\{AB, BC\}$
 - (b) $\{AG, CF\}$
 - (c) $\{AG, HF\}$
 - (d) $\{CD, DE\}$
 - (e) $\{DF, HF\}$

PART I

21. Which of the following sets contain only back edges?

- (a) $\{AB, BC\}$
- (b) $\{AG, CF\}$
- (c) $\{AG, HF\}$
- (d) $\{CA, GC\}$
- (e) $\{DE, DF\}$

22. Which of the following sets contain only forward edges?

- (a) $\{AB, BC\}$
- (b) $\{AG, CF\}$
- (c) $\{AG, HF\}$
- (d) $\{CD, DE\}$
- (e) $\{DF, HF\}$

23. Which of the following is/are cross edges?

- (a) AB
- (b) CA
- (c) CF
- (d) GC
- (e) HF

24. What strongly connected components (SCC) are identified using Kosarajus's algorithm on the graph in Figure 35?

- (a) A-B-C
- (b) A-C-G
- (c) C-D-F
- (d) A-B-C-G
- (e) A-B-C-D-F-G

PART I

25. How many strongly connected components are identified in the graph in Figure 35 according to the Tarjan's algorithm?
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
 - (e) 6
26. Which of the following statements is/are true regarding balanced trees and the balanced tree algorithms?
- (a) If the choice is between an AVL tree and the Red-Black tree, the AVL tree is the tree of choice if there are a large (10^4) amount of insertions and deletions.
 - (b) If the choice is between the Red-Black tree and the DSW algorithm for a tree which only has inserts and is used for search after all inserts are complete, both methods are equally efficient.
 - (c) Out of the three tree balancing techniques, namely, the DSW algorithm, Red-Black tree, and AVL tree, only the DSW algorithm will guarantee a complete binary tree.
 - (d) The main requirement of any tree balancing algorithm is to ensure a $O(\log n)$ worst-case search time complexity or a search time complexity closer to $O(\log n)$.
 - (e) The DSW algorithm is the most efficient tree-balancing algorithm when there are many arbitrary insertions and deletions.
27. Which of the following statements is/are true regarding Red-Black trees?
- (a) Red-Black trees are more balanced than the AVL trees.
 - (b) Red-Black trees perform comparatively less balancing than the AVL trees for the same set of operations.
 - (c) For any given subtree of a Red-Black tree, the black height property has to be preserved.
 - (d) Red-Black trees have four (4) deletion cases in which one (1) case falls through to others.
 - (e) When an extra black is passed up the tree in a Red-Black tree, the extra black can be removed when it reaches the root since it will not damage the black height property.

PART I

Consider the code snippet below and answer **Questions 28 to 30**.

```
p, i = 0;
for (i = 1; i < n; i = i*2)  <= Loop A
{
    p++;
}
for (i = n; i > 1; i = i/2){ ... }  <= Loop B
for (i = 1; i < p; i = i*2){ ... }  <= Loop C
```

28. What is the running time complexity of Loop A?

- (a) $\Theta(n)$
- (b) $\Theta(\sqrt{n})$
- (c) $O(\log \sqrt{n})$
- (d) $\lceil O(\log(n)) \rceil$
- (e) $\lfloor O(\log(n)) \rfloor$

29. What is the running time complexity of Loop B?

- (a) $\Theta(n)$
- (b) $\Theta(\sqrt{n})$
- (c) $O(\log \sqrt{n})$
- (d) $\lceil O(\log(n)) \rceil$
- (e) $\lfloor O(\log(n)) \rfloor$

30. What is the running time complexity of Loop C?

- (a) $O(\sqrt{n})$
- (b) $O(n)$
- (c) $O(\log \sqrt{n})$
- (d) $O(\log \log(n))$
- (e) $O(\log(n^3))$

31. Solve the recurrence relation $T(n) = 7T(n/2) + 3n^2 + 2$?

- (a) $O(n^{0.5})$
- (b) $O(n^{1.6})$
- (c) $O(n^2)$
- (d) $O(n^{2.4})$
- (e) $\Theta(n^{2.8})$

PART I

32. Master theorem can be applied on which of the following recurrence relation?

- (a) $T(n) = 2T(n/2) + 2^n$
- (b) $T(n) = 2T(n/3) + \sin(n)$
- (c) $T(n) = T(n - 2) + 2n^2 + 1$
- (d) $T(n) = T(n/6) + 3^n$
- (e) $T(n) = 3T(n/2 + 47) + 2n^2 + 10 * n^{-1/2}$

33. How many times would the loop *for*($i = 0; i < n; i++$) execute?

- (a) $n - 1$ times
- (b) n times
- (c) $n + 1$ times
- (d) $n + 2$ times
- (e) $n + 3$ times

34. What is the space complexity of matrix multiplication?

- (a) $O(n)$
- (b) $O(n^2)$
- (c) $O(n^3)$
- (d) $\Theta(n^2)$
- (e) $\Theta(1)$

PART II

- Consider the following scenario of a graph G of n nodes and answer Questions 1 to 5.

Spiral is a computer network that spans the world. The spiral network has a central node that connects to other nodes in the network. The network contains m ($m > 0$) layers around the centre, each with $4*m$ nodes. The central node is to be considered layer zero. There is a predefined set of links between each layer. The pattern of links is that there will be α links between layer $m - 1$ and m .

1. Draw the Spiral network as explained above. The drawing should have at least two (2) layers in addition to layer zero. **[05 marks]**
2. What is the total number of nodes in the Spiral? **[05 marks]**
3. How many links are between the layers of the Spiral if it has 465 nodes in the layers? Clearly state your assumptions. **[05 marks]**
4. Considering n to be large ($> 10^6$), explain what the optimal search algorithm to search for a given computer (X) in the Spiral is. Your answer should consider all possible placements of X in the graph. **[15 marks]**

———— *End of Paper* ————